



Department of Computer Science
COMP2421 (Fall Semester 2025/2026)

Project#1 Due Date: Sunday (30 November 2025)

In this project, you will implement a simulator an airport's air traffic control (ATS) to handle and schedule the departure and arrival of airplanes. Planes can have the following states: arrival, departure, and emergency (for arrival only), and each should be handled in a separate data structure based on the order they were entered. As well, departed or cancelled flights should be removed but not deleted, and should be maintained in another data structure with their status. Deleted means the data will be deleted permanently, while removed means the data, a plane, is no longer in the departure/arrival data but its information are kept in another data structure.

Your project should do the following tasks through a proper menu:

1. Read two files: the first called flights.txt; containing the information of the flights, and the second called passengers.txt that contains passengers' information.

flight.txt

flight ID; state; date; time

AMM220; Arrival; 20-12-2025; 23:00

SNG456; Departure; 20-12-2025; 23:50

passenger.txt

Name, passport number, flight ID

Ahmed; 123123; AMM220

Once you read the files, you will create a list such that each plane is a type of list and each node is a passenger. As well, you should create the structures responsible for the departure, arrival, and emergency.

2. Print flights: this option will print all flights: arrival, departure, and emergency.
3. Print flight detail: print the details of any flight and all passenger information given the flight number.
4. Add a new flight: this option will allow users to input a new flight to the pipeline. There are two types of flights to be added: normal and emergency flights. Emergency flights should be handled first.
5. Land a flight: lands next flight. If there are emergency flights, then land them first. When landing a plane, print the flight information.
6. Depart plane: depart planes from the airport.
7. Cancel flight: any cancelled flight should be removed but not deleted from the system.
8. Display flight status: this option will print all planes that are ready to land/depart, all emergency planes, and all departed/cancelled planes. Only print the flight information, including total count of passengers, without detailed passenger information.
9. Manage passengers:
 - a. Print passengers of a given flight.
 - b. Add a passenger to a flight.
 - c. Remove a passenger from a flight.
10. Exit. sub menu

Simulation of the program:

Suppose the input file *flight.txt* contains the following inputs:

```
AMM100; Arrival; 20-12-2025; 23:00
SNG200; Departure; 20-12-2025; 23:50
AMM300; Emergency; 20-12-2025; 23:55
```

And the user inputs the following sequence:

```
Option (5) Land a plane
Option (6) Depart a plane
```

Then the program should output the following:

```
AMM300 Emergency landed
SNG200 Departed
```

Here, flight AMM300 has landed even the request is to land only one plane. This is because emergency planes should be handled before you handle flight number AMM100 .

The deadline of this assignment is on Sunday 30 November 2025 before 8:00PM. Any late submission after 8:00PM to the midnight will be penalized 30%, and after will not be accepted. Please make sure that your application is running properly on your laptop before the lecture. Project discussions will take place in the lecture room.

Grading policy:

1. Your application should have all functionalities working properly. **Twenty** marks will be graded for the functionality of the project;
2. The following notes will make up the remaining 10 marks of the grade:
 - a. There has to be adequate documentation and comments in the code (i.e., functions, loops, etc.);
 - b. Your code should follow the code convention (i.e., spaces, indentations, etc.); and
 - c. Your application should contain a menu to allow the user to select which option (s) he would like to run.

Notes and submission instructions:

1. **This is individual work.** It should represent your own efforts. It is fine to discuss your work and to ask your colleagues, but you are not allowed to copy/paste the work of others or give your work to anyone else. You are not allowed to post/copy from other websites and/or social media and this will be considered as cheating. Using AI tools to assist writing the code will result in zero grade.
2. Any **plagiarized** code will not be marked.
3. **Document format.** Please submit only the code file (**c** file) containing the code of your project. Please rename it as follows: **"P1_YourStudentID_FirstNameLastName_SectionNo.c"**.
4. **Input/output file name.** Make sure that the input/output file names are the same as in the specifications.
5. Include your full name, student ID, and section number in the beginning of your file.
6. Please do not compress the file, only the C-file is needed.
7. Files not following the convention in point 2 will not be marked.

Good luck!